Name: ______________________

WORKSHEET

Functions and relations

The hottest places in Hell are reserved for those who …

11 15   12 11 16 8 13   10 5   3 4 8 9 12

16 10 4 9 6   14 4 11 13 8 13   16 9 11 15 12 9 11 15

12 1 8 11 4   15 8 2 12 4 9 6 11 12 7

Dante Alighieri

Rules

A **relation** is a set of ordered pairs, (x, y). A **function** is a relation where each input value, x, has at most one output value, y.

A vertical line test can be used to determine whether a graph is a function. If a vertical line can be passed over the graph and never passes through more than one point at a time, then the graph is a function.

Questions

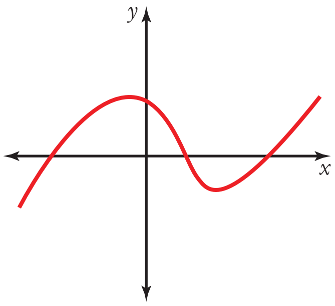
Determine whether the following relations are functions.

1 (1, 3), (2, 5), (5, 11), (9, 19)

2 $y = 7 - 4x$

3 $(-2, 1), (3, 4), (1, 5), (6, 2), (3, 8)$

4

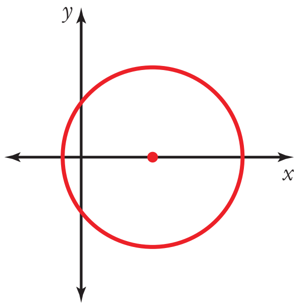


5 $y^2 = x$

6 $(x, 3x - 2)$

7 $(-4, 1), (5, 3), (-1, 4), (-4, 2), (2, 3)$

8



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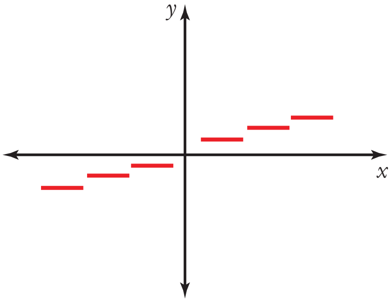
9 $y = 3x^2 - x + 7$

10

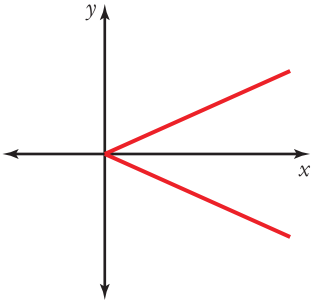


11 $(x, 1 \pm x)$

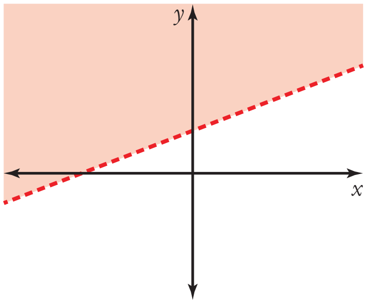
12 $(-3, 4), (0, 4), (5, 4), (2, 4)$

13 $y = (x + 1)(x - 3)(x + 2)$

14



15



16 $y^2 = 5x - 1, x \geq 2$

Solutions

Question	Function	Not a function
1	H	N
2	U	K
3	V	G
4	R	P
5	J	F
6	L	B
7	Q	Y
8	C	E
9	A	Z
10	O	D
11	X	I
12	T	P
13	S	K
14	V	C
15	W	N
16	B	M

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